Feiyang Liu

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Research Interests

Language-Centric Multimodal Systems: Scalable, Unified, and Controllable Multimodal Models; Unified Multimodal Pretraining; Solving Vision Problems with LLMs; Symbolic Vision; End-to-End Multimodal Training Systems.

Education

South China University of Technology Sep 2022 – Jun 2027 (expected)
B.Eng. in Intelligent Science and Technology

University of California, Berkeley Aug 2024 – May 2025
Visiting Student

Publications

1. **PixelBench: Can LLMs Understand and Generate Images Pixel by Pixel?**
   Haoquan Zhang, **Feiyang Liu**, Juanxi Tian, Wanhua Li, Haodong Duan, Weiyang Liu *2026* · [Project] [Code] [Models]

2. **Confidence-Guided Multi-Scale Aggregation for Sparse-View High-Resolution 3D Gaussian Splatting**
   Qinzheng Zhou, **Feiyang Liu**, Lijing Lu, Zhihang Li *CVPR 2026*, **Highlight** · [Paper] [Poster]

3. **Commit Only When Justified: Counterfactual Topology Selection for Sparse-View 3DGS**
   Qinzheng Zhou, **Feiyang Liu**, Lijing Lu, Zhihang Li *Under review (Phase 2), AAAI 2027*

Research Experience

End-to-End Training of Unified Multimodal Models (Ongoing Project) Ongoing
Project Lead; UC Santa Cruz and Gemini Team

- Leading an end-to-end empirical study (in the spirit of *Towards Physics of Multimodal Pretraining* and *Beyond Language Modeling*) on **large-scale unified pretraining** and **multi-stage post-training** of Unified Multimodal Models (UMMs) across language modeling, visual understanding, and image generation (details to be shared upon release).
- Built and optimized a modular JAX/Flax pretraining, post-training, and decontaminated evaluation stack across **hundreds of TPU v5p/v6e chips** (multi-host/multi-slice sharding, ZeRO-1, rematerialization, elastic recovery, XLA profiling).

Visual Perception with Modern LLMs (Ongoing Project) Ongoing
Research Contributor; CUHK and ByteDance Seed

- Collaborating on an ongoing empirical study exploring how modern LLMs tackle core visual perception tasks (details to be shared upon release).

PixelBench
Research Contributor; CUHK and ByteDance Seed

- Probed LLMs as general visual generators without agentic tools, visual tokenizers, or diffusion modules, generating pixel images, 3D voxels, and video in text as **symbolic visual primitives** and scaling with coding/reasoning ability.
- Co-developed **Pixel Reconstruction Tuning (PRT)** in pretraining/post-training using text-encoded pixels to boost visual understanding; analyzed visual abstraction (downsampling, resolution) and built the workbench.

Sparse-View 3D Reconstruction 2025 – 2026
Research Collaborator; Huazhong University of Science and Technology

- Designed confidence-guided multi-scale aggregation and counterfactual topology selection algorithms to mitigate high-frequency floaters and premature topological lock-in in sparse-view 3D Gaussian Splatting.

Technical Skills

- **ML Systems:** JAX/Flax, PyTorch, multi-host/multi-slice TPU training, sharding, ZeRO-1, XLA profiling.
- **Graphics & 3D:** C++ graphics programming, path tracing, mesh processing, physical simulation, 3DGS; Three.js / React Three Fiber, Blender, OpenGL/GLSL, Unity.
- **Programming:** Python, C/C++, Java.